DSA FAQ (Experiments 6 to 9)

Experiment 6: Queue (and Stack with Linked List)

1. **What is a queue?**
   A linear data structure (FIFO) where insertion happens at the rear and deletion at the front.
2. **How is a queue represented?**
   Using an array or a linked list.
3. **Types of queues?**
   Linear queue, circular queue, priority queue, deque.
4. **Concept of circular queue?**
   The last position is connected back to the first, so the rear wraps around (rear = (rear+1) % MAX). This reuses freed space.
5. **Applications of queues?**
   CPU and disk scheduling, printer spooling, buffering (such as the UART buffer in Exp 10), BFS, queue simulation.
6. **What is a priority queue?**
   Each element has a priority. The higher-priority element is processed first; elements with equal priority are processed in the order they arrived.
7. **What is a deque?**
   A double-ended queue. Elements can be inserted or removed at both the front and the rear, but not in the middle.

Experiment 7: Postfix Evaluation

1. **Types of expressions?**
   Infix, prefix and postfix.
2. **Examples of each?**
   Infix: a+b. Prefix: +ab. Postfix: ab+.
3. **Data structure used for evaluation?**
   Stack.
4. **Infix a+b+c+d to prefix?**
   +++abcd
5. **Infix a+b+c+d to postfix?**
   ab+c+d+

Experiment 8: Binary Search Tree (Inorder Traversal)

No FAQs are given in the manual for this experiment.

Experiment 9: Trees (as given in the manual)

1. **What is a tree?**
   A non-linear data structure that stores hierarchical data as nodes connected by edges, with one root node.
2. **What is a binary tree?**
   A tree that is empty, or consists of a root and two disjoint binary trees called the left and right subtrees. Each node has at most two children.

3. **Types of binary trees?**
   Full, complete, skewed, binary search tree, threaded, AVL and expression trees.
4. **Tree traversal techniques?**
   Recursive and non-recursive (iterative).
5. **Recursive traversal techniques?**
   Inorder (Left, Root, Right), Preorder (Root, Left, Right), Postorder (Left, Right, Root).
6. **What are AVL trees?**
   Self-balancing binary search trees in which the height difference between the left and right subtrees of every node is at most 1.
7. **Basic operations on a tree?**
   Create (insert), search, delete, traversal.
8. **Degree and height of a tree?**
   The degree of a node is its number of subtrees (children). The height or depth of a tree is the maximum level of any node.
9. **Representation of a tree?**
   - Linked list: each node is struct node { int data; struct node *left, *right; };
   - Array: root at index 1. For a node at index i, left child is at 2i, right child at 2i+1, parent at i/2. Missing children leave empty slots.